Exp No:11 Experiment to understand panda library use cases in Data science.

AIM:

To understand panda library use cases in Data science.

Description:

Understand data frame use cases using panda library

CODE:

```
import numpy as np
import pandas as pd
list1 = [
   [1, "Arun", 85],
   [2, "Priya", 92]
]
df = pd.DataFrame(list1)
df
df.columns = ["ID", "Name", "Marks"]
df
df.info()
df = pd.read_csv("student.csv")
df
df.info()
df.head()
df.tail()
df["Marks"]
type(df["Marks"])
df["Marks"].mean()
df["Marks"].median()
df["Marks"].mode()
df["Marks"].var()
```

```
df.describe()

df.describe(include='all')

empcol = df.columns

empcol

emparray = df.values

emparray

student_df = pd.DataFrame(
   emparray,
   columns=empcol
)

student_df
```

OUTPUT:

	0	1	2
0	1	Arun	85
1	2	Priya	92

	Name	Age	Marks	Department
3	Divya	20	88	EEE
4	Kiran	19	95	CSE
5	Meena	20	81	ECE
6	Ajay	19	73	EEE
7	Sneha	20	90	CSE

	ID	Name	Marks
0	1	Arun	85
1	2	Priya	92

	Name	Age	Marks	Department
0	Arun	19	85	CSE
1	Priya	20	92	CSE
2	Ravi	19	76	ECE
3	Divya	20	88	EEE
4	Kiran	19	95	CSE
5	Meena	20	81	ECE
6	Ajay	19	73	EEE
7	Sneha	20	90	CSE

	Name	Age	Marks	Department
0	Arun	19	85	CSE
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<class 'pandas.core.frame.DataFrame'>

RangeIndex: 8 entries, 0 to 7

Data columns (total 4 columns):

#	Column	Non-Null Count	Dtype
0	Name	8 non-null	object
1	Age	8 non-null	int64
2	Marks	8 non-null	int64
3	Department	8 non-null	object

dtypes: int64(2), object(2)

memory usage: 388.0+ bytes

np.float64(85.0)

86.5

	Marks
0	73
1	76
2	81
3	85
4	88
5	90
6	92
7	95

dtype: int64

60.57142857142857

7.782764841072135

	Age	Marks
count	8.000000	8.000000
mean	19.500000	85.000000
std	0.534522	7.782765
min	19.000000	73.000000
25%	19.000000	79.750000
50%	19.500000	86.500000
75%	20.000000	90.500000
max	20.000000	95.000000

	Name	Age	Marks	Department
count	8	8.000000	8.000000	8
unique	8	NaN	NaN	3
top	Arun	NaN	NaN	CSE
freq	1	NaN	NaN	4
mean	NaN	19.500000	85.000000	NaN
std	NaN	0.534522	7.782765	NaN
min	NaN	19.000000	73.000000	NaN
25%	NaN	19.000000	79.750000	NaN
50%	NaN	19.500000	86.500000	NaN
75%	NaN	20.000000	90.500000	NaN
max	NaN	20.000000	95.000000	NaN

Index(['Name', 'Age', 'Marks', 'Department'], dtype='object')

	Name	Age	Marks	Department
0	Arun	19	85	CSE
1	Priya	20	92	CSE
2	Ravi	19	76	ECE
3	Divya	20	88	EEE
4	Kiran	19	95	CSE
5	Meena	20	81	ECE
6	Ajay	19	73	EEE
7	Sneha	20	90	CSE

RESULT:

The pandas library was successfully used to create load and explore a data frame.various dataframe operations and statistical functions were performed successfully of the student datset.